

Unit 5

System of First Order Linear Differential Equation

Introduction

We have so far solved ordinary differential equations involving only one dependent variable like $y' = \frac{dy}{dx} = f(x, y)$, $y'' = \frac{d^2y}{dx^2} = f(x, y, y')$, ... , $y^{(n)} = \frac{d^ny}{dx^n} = f(x, y, y', y'', \dots, y^{(n-1)})$. There are many applications or problems which require two or more dependent variables each of which is a function of a single independent variable t (specially time). Also, the dependent variables may be interrelated. Such problems in science and social sciences lead to the system of simultaneous ordinary differential equations. Notationally, we use t for independent variable and the dependent variables are denoted by $x_1, x_2, x_3, \dots, x_n$ (which are unknown functions of t) and primes are used to denote their derivatives with respect to t . For example

$$\begin{aligned}x'_1 &= f_1(t, x_1, x_2, x_3, \dots, x_n) \\x'_2 &= f_2(t, x_1, x_2, x_3, \dots, x_n) \\&\dots \dots \dots \dots \dots \dots \dots \\&\dots \dots \dots \dots \dots \dots \dots \\x'_n &= f_n(t, x_1, x_2, x_3, \dots, x_n)\end{aligned}$$

Is a system of ordinary first order differential equation with n unknown functions $x_1, x_2, x_3, \dots, x_n$ of a dependent variable t .

Linear and Non-linear System

The system

$$\begin{aligned}x'_1 &= f_1(t, x_1, x_2, x_3, \dots, x_n) \\x'_2 &= f_2(t, x_1, x_2, x_3, \dots, x_n) \\&\dots \dots \dots \dots \dots \dots \dots \\&\dots \dots \dots \dots \dots \dots \dots \\x'_n &= f_n(t, x_1, x_2, x_3, \dots, x_n)\end{aligned}$$

is said to be linear if the functions $f_1, f_2, \dots, f_n$ are linear in the variables $x_1, x_2, x_3, \dots, x_n$. And if the system is not linear then it is called non-linear.

Transformation of Higher Order Differential Equation into a Linear System and Vice Versa.

A higher order differential equation can be transformed into a first linear system and a system of first order linear differential equation can similarly be transformed into a higher order differential equation. This is illustrated in the following examples.

Theorem 1 (Statement Only)

If each of the functions $F_1, F_2, \dots, F_n$ and the partial derivatives

$\frac{\partial F_1}{\partial x_1}, \frac{\partial F_1}{\partial x_2}, \dots, \frac{\partial F_1}{\partial x_n}; \frac{\partial F_2}{\partial x_1}, \frac{\partial F_2}{\partial x_2}, \dots, \frac{\partial F_2}{\partial x_n}; \dots, \frac{\partial F_n}{\partial x_1}, \frac{\partial F_n}{\partial x_2}, \dots, \frac{\partial F_n}{\partial x_n}$ are continuous in a region R of $tx_1x_2 \dots x_n$ space defined by $\alpha < t < \beta, \alpha_1 < x_1 < \beta_1, \dots, \alpha_n < x_n < \beta_n$ and of the point $(t_0, x_1^0, x_2^0, \dots, x_n^0)$ is in R , then there is an interval $|t - t_0| < \delta$ in which there exists a unique solution

$x_1 = \phi_1(t), x_2 = \phi_2(t), \dots, x_n = \phi_n(t)$ of the system of differential equation

$$\begin{aligned}x'_1 &= F_1(t, x_1, x_2, \dots, x_n) \\x'_2 &= F_2(t, x_1, x_2, \dots, x_n) \\&\dots \dots \dots \dots \dots \dots \dots \\&\dots \dots \dots \dots \dots \dots \dots\end{aligned}$$

$$x'_n = F_n(t, x_1, x_2, \dots, x_n)$$

That also satisfies the initial conditions

$$x_1(t_0) = x_1^0, x_2(t_0) = x_2^0, \dots, x_n(t_0) = x_n^0.$$

Theorem 2 (Statement Only) Let the functions $p_{11}, p_{12}, \dots, p_{1n}; p_{21}, p_{22}, \dots, p_{2n}; \dots; p_{n1}, p_{n2}, \dots, p_{nn}; g_1, g_2, \dots, g_n$ be continuous in an open interval $I: \alpha < t < \beta$. Then, there exists a unique solution $x_1 = \phi_1(t), x_2 = \phi_2(t), \dots, x_n = \phi_n(t)$ of the system

$$\begin{aligned} x'_1 &= p_{11}(t)x_1 + p_{12}(t)x_2 + \dots + p_{1n}(t)x_n + g_1(t) \\ x'_2 &= p_{21}(t)x_1 + p_{22}(t)x_2 + \dots + p_{2n}(t)x_n + g_2(t) \\ &\dots \dots \dots \dots \dots \dots \dots \dots \dots \dots \\ &\dots \dots \dots \dots \dots \dots \dots \dots \dots \dots \\ x'_n &= p_{n1}(t)x_1 + p_{n2}(t)x_2 + \dots + p_{nn}(t)x_n + g_n(t) \end{aligned}$$

That also satisfies the initial conditions

$$x_1(t_0) = x_1^0, x_2(t_0) = x_2^0, \dots, x_n(t_0) = x_n^0.$$

where t_0 is any point in I and $x_1^0, x_2^0, \dots, x_n^0$ are any prescribed numbers. Moreover, the solution exists throughout the interval I .

Examples 1 Transform the following equations into system of first order equations

- i. $u'' + 0.5u' + 2u = 3\sin t$
- ii. $t^2u'' + tu' + (t^2 - 1)u = 0$

Solution i:

The given ordinary differential equation is:

$$u'' + 0.5u' + 2u = 3\sin t \quad \dots (1)$$

Put $x_1 = u$ and $x_2 = u'$

$$\therefore x'_1 = u' = x_2$$

and

$$\begin{aligned} x'_2 &= u'' = 3\sin t - 0.5u' - 2u, \text{ using (1)} \\ &\Rightarrow x'_2 = 3\sin t - 0.5x_2 - 2x_1 \end{aligned}$$

Hence, the required system of first order linear differential equation is

$$\begin{aligned} x'_1 &= x_2 \\ x'_2 &= 3\sin t - 0.5x_2 - 2x_1 \end{aligned}$$

Example 2 Transform the initial value problem into an initial value problem for two first order equations.

$$u'' + p(t)u' + q(t)u = g(t), \quad u(0) = u_0, \quad u'(0) = u'_0.$$

Solution:

The given ODE is

$$u'' + p(t)u' + q(t)u = g(t) \quad \dots (1)$$

with initial conditions

$$u(0) = u_0, \quad u'(0) = u'_0.$$

Let $x_1 = u$ and $x_2 = u'$.

$$\therefore x'_1 = u' = x_2$$

And $x'_2 = u'' = g(t) - p(t)u' - q(t)u$

$$g(t) - p(t)x_2 - q(t)x_1$$

Therefore, the required system is

$$\begin{aligned} x'_1 &= x_2 \\ x'_2 &= -p(t)x_2 - q(t)x_1 + g(t) \end{aligned}$$

Also, using $u(0) = u_0$, we get $x_1(0) = u_0$

and using $u'(0) = u'_0$, we get $x_2(0) = u'_0$

Hence, the required IVP for two first order equations is

$$x'_1 = x_2, x_1(0) = u_0$$

$$x'_2 = -p(t)x_2 - q(t)x_1 + g(t), x_2(0) = u'_0$$

Example 3 Consider the system

$$x'_1 = x_1 - 2x_2, x_1(0) = -1$$

$$x'_2 = 3x_1 - 4x_2, x_2(0) = 2$$

a. Transform the above system into a single equation of second order.

b. Find x_1 and x_2 that also satisfy the given initial conditions.

Solution

The given system is

$$x'_1 = x_1 - 2x_2, x_1(0) = -1 \quad \dots \quad (1)$$

$$x'_2 = 3x_1 - 4x_2, x_2(0) = 2 \quad \dots \quad (2)$$

From equation (1)

$$x'_1 = x_1 - 2x_2$$

$$\Rightarrow 2x_2 = -x'_1 + x_1$$

$$\Rightarrow x_2 = -\frac{1}{2}x'_1 + \frac{1}{2}x_1 \quad \dots \quad (3)$$

$$\Rightarrow x'_2 = -\frac{1}{2}x''_1 + \frac{1}{2}x'_1 \quad \dots \quad (4)$$

From (2) and (4)

$$-\frac{1}{2}x''_1 + \frac{1}{2}x'_1 = 3x_1 - 4x_2 = 3x_1 - 4[-\frac{1}{2}x'_1 + \frac{1}{2}x_1], \text{ using (3)}$$

$$\Rightarrow -\frac{1}{2}x''_1 + \frac{1}{2}x'_1 = 3x_1 + 2x'_1 - 2x_1$$

$$\Rightarrow x''_1 - x'_1 = -6x_1 - 4x'_1 + 4x_1$$

$$\Rightarrow x''_1 + 3x'_1 + 2x_1 = 0 \quad \dots \quad (5)$$

which is required second order homogeneous differential equation with constant coefficient.

It's A.E. is

$$m^2 + 3m + 2 = 0$$

$$\therefore m = -1, -2$$

Hence, the general solution of (5) is

$$x_1(t) = c_1 e^{-t} + c_2 e^{-2t} \quad \dots \quad (6)$$

$$\Rightarrow x'_1(t) = -c_1 e^{-t} - 2c_2 e^{-2t}$$

Now, from (3)

$$\Rightarrow x_2(t) = -\frac{1}{2}x'_1(t) + \frac{1}{2}x_1(t)$$

$$= -\frac{1}{2}[-c_1 e^{-t} - 2c_2 e^{-2t}] + \frac{1}{2}[c_1 e^{-t} + c_2 e^{-2t}]$$

$$\Rightarrow x_2(t) = c_1 e^{-t} + \frac{3}{2}c_2 e^{-2t} \quad \dots \quad (7)$$

Using $x_1(0) = -1$, (6) gives

$$c_1 + c_2 = -1 \quad \dots \quad (8)$$

Using $x_2(0) = 2$, (7) implies

$$2 = c_1 + \frac{3}{2}c_2 \Rightarrow 2c_1 + 3c_2 = 4 \quad \dots \quad (9)$$

Solving (8) and (9), we get

$$c_1 = -7, c_2 = 6$$

Hence, the required solution is

$$x_1(t) = -7e^{-t} + 6e^{-2t}$$

$$\text{and } x_2(t) = -7e^{-t} + 9e^{-2t}$$

Example 4 Transform the given initial value problem (IVP) into an initial value problem for two first order linear equations:

$$u'' + u' + 4u = 2 \sin 4t; u(0) = 1, u'(0) = 3$$

Solution

The given IVP is

$$u'' + u' + 4u = 2 \sin 4t \quad \dots (1)$$

with initial conditions

$$u(0) = 1, u'(0) = 3$$

Let $x_1 = u$ and $x_2 = u'$.

$$\therefore x_1' = u' = x_2$$

and $x_2' = u'' = -u' - 4u + 2 \sin 4t$

$$\text{or, } x_2' = -x_2 - 4x_1 + 2 \sin 4t$$

Also, $u(0) = 1 \Rightarrow x_1(0) = 1$

and $u'(0) = 3 \Rightarrow x_2(0) = 3$

Hence, the required IVP for two first order linear equations is

$$x_1' = x_2; x_1(0) = 1$$

$$x_2' = -4x_1 - x_2 + 2 \sin 4t; x_2(0) = 3$$

Review of Matrices

$$\begin{array}{ccccccccc} \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \end{array}$$

Matrix of Functions

In some specific purposes, the components of a vector or a matrix are functions of a scalar variable, in particular,

$$x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}$$

$$A(t) = \begin{bmatrix} a_{11}(t) & a_{12}(t) & \dots & a_{1n}(t) \\ a_{21}(t) & a_{22}(t) & \dots & a_{2n}(t) \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1}(t) & a_{n2}(t) & \dots & a_{nn}(t) \end{bmatrix}$$

If each $a_{ij}(t)$ is continuous at $t = t_0$ on an interval $\alpha < t < \beta$, then A is said to be continuous at $t = t_0$. If each $a_{ij}(t)$ is differentiable, then $A(t)$ is also differentiable. Moreover, the derivative of $A(t)$ is $\frac{dA}{dt} = \left(\frac{da_{ij}(t)}{dt} \right)$.

If A and B are matrices of functions of appropriate order and C is a matrix of constants, then we have

$$i. \quad \frac{d}{dt}(CA) = C \frac{dA}{dt}$$

$$ii. \quad \frac{d}{dt}(A \pm B) = \frac{dA}{dt} \pm \frac{dB}{dt}$$

$$iii. \frac{d}{dt}(A \times B) = A \frac{dB}{dt} + B \frac{dA}{dt}$$

Similarly,

$$i. \int_a^b A(t)dt = \left[\int_a^b a_{ij}(t)dt \right]$$

Example

Let $A(t) = \begin{bmatrix} t^2 & 2t \\ 1 & e^t \end{bmatrix}$. Then,

$$\int_1^2 A(t)dt = \begin{bmatrix} \int_1^2 t^2 dt & \int_1^2 2t dt \\ \int_1^2 1. dt & \int_1^2 e^t dt \end{bmatrix} = \begin{bmatrix} \frac{7}{3} & 3 \\ 1 & e(e-1) \end{bmatrix}$$

System of Linear Algebraic Equations

Consider the following system of n simultaneous linear algebraic equation in n variables.

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ \dots & \dots \dots \dots \dots \dots \dots \dots \dots \\ a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n &= b_n \end{aligned}$$

This system can be written in matrix form as

$$Ax = b \quad \dots \quad (1),$$

$$\text{where, } A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}, x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \text{ and } b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

If $b = 0$, then the system become $Ax = 0$, which is called homogeneous system, otherwise (i.e. if $b \neq 0$) it is called non-homogeneous. If $|A| \neq 0$, i.e. A is non-singular, then there is a unique solution of the system. Since A^{-1} exists when $|A| \neq 0$, the solution of (1) can be found by

$$x = A^{-1}b \quad \dots \quad (2)$$

On the other hand, if A is singular i.e. if $|A| = 0$, then the system is inconsistent (with no solution) or is consistent with infinitely many solutions.

The homogeneous system $Ax = 0$ may have infinitely many solutions in addition to the trivial

solution $\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$, which are called the non-trivial solutions.

The augmented matrix of the system (1) is

$$[A: b] = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & : & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & : & b_2 \\ \vdots & \vdots & \ddots & \vdots & : & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} & : & b_n \end{bmatrix}$$

Theorem

If A is an $n \times n$ matrix, then the homogeneous system $Ax = 0$ has exactly one (unique) solution, (called the trivial solution) if and only if A is invertible. It will have infinitely many solutions (called the non-trivial solution) if A is not invertible.

Solution of Linear System

1. If the augmented matrix

$$[A: b] = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & : & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & : & b_2 \\ \vdots & \vdots & \vdots & \vdots & : & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & : & b_n \end{bmatrix}$$

can be reduced into an ECHELON form of the form

$$\begin{bmatrix} c_{11} & c_{12} & c_{13} & \cdots & c_{1n} & : & d_1 \\ 0 & c_{22} & c_{23} & \cdots & c_{2n} & : & d_2 \\ 0 & 0 & c_{33} & \cdots & c_{3n} & : & d_3 \\ \vdots & \vdots & \vdots & \vdots & \vdots & : & \vdots \\ 0 & 0 & 0 & \cdots & c_{nn} & : & d_n \end{bmatrix},$$

then the associated system of equations has a unique solution. In this case all the unknown variables are BASIC variables.

2. If the augmented matrix

$$[A: b] = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & : & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & : & b_2 \\ \vdots & \vdots & \vdots & \vdots & : & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & : & b_n \end{bmatrix}$$

can be reduced into an ECHELON form of the form

$$\begin{bmatrix} c_{11} & c_{12} & c_{13} & \cdots & c_{1n} & : & d_1 \\ 0 & c_{22} & c_{23} & \cdots & c_{2n} & : & d_2 \\ 0 & 0 & c_{33} & \cdots & c_{3n} & : & d_3 \\ \vdots & \vdots & \vdots & \vdots & \vdots & : & \vdots \\ 0 & 0 & 0 & \cdots & 0 & : & 0 \end{bmatrix},$$

then the associated system of equations has infinitely many solutions. In such a case, at least one variable x_n is a FREE variable.

3. If the augmented matrix

$$[A: b] = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & : & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & : & b_2 \\ \vdots & \vdots & \vdots & \vdots & : & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & : & b_n \end{bmatrix}$$

can be reduced into an ECHELON form of the form

$$\begin{bmatrix} c_{11} & c_{12} & c_{13} & \cdots & c_{1n} & : & d_1 \\ 0 & c_{22} & c_{23} & \cdots & c_{2n} & : & d_2 \\ 0 & 0 & c_{33} & \cdots & c_{3n} & : & d_3 \\ \vdots & \vdots & \vdots & \vdots & \vdots & : & \vdots \\ 0 & 0 & 0 & \cdots & 0 & : & d_n \end{bmatrix},$$

then the associated system of equations has no solutions. In such a case, $0 \cdot x_n = d_n \neq 0$ is not possible for any value of x_n .

Example 1 Solve the following system

$$x_1 + 2x_2 - x_3 = 0$$

$$2x_1 + x_2 - x_3 = 0$$

$$x_1 - x_2 - 2x_3 = 0$$

Solution

The augmented matrix is

$$\begin{bmatrix} 1 & 2 & -1 & : & 0 \\ 2 & 1 & 1 & : & 0 \\ 1 & -1 & 2 & : & 0 \end{bmatrix} \approx \begin{bmatrix} 1 & 2 & -1 & : & 0 \\ 0 & -3 & 3 & : & 0 \\ 0 & -3 & 3 & : & 0 \end{bmatrix}, (why?)$$

$$\approx \begin{bmatrix} 1 & 2 & -1 & : & 0 \\ 0 & -3 & 3 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}, (why?)$$

This shows that the system has infinite solution (why?).

Example 2 Solve the following system

$$\begin{aligned} x_1 - x_3 &= 0 \\ 3x_1 + x_2 + x_3 &= 1 \\ -x_1 + x_2 + 2x_3 &= 2 \end{aligned}$$

Solution

The given system is

$$\begin{aligned} x_1 - x_3 &= 0 \\ 3x_1 + x_2 + x_3 &= 1 \\ -x_1 + x_2 + 2x_3 &= 2 \end{aligned}$$

The augmented matrix is

$$\begin{aligned} &\begin{bmatrix} 1 & 0 & -1 & : & 0 \\ 3 & 1 & 1 & : & 1 \\ -1 & 1 & 2 & : & 2 \end{bmatrix} \\ &\approx \begin{bmatrix} 1 & 0 & -1 & : & 0 \\ 0 & 1 & 4 & : & 1 \\ 0 & 1 & 1 & : & 2 \end{bmatrix}, (???) \\ &\approx \begin{bmatrix} 1 & 0 & -1 & : & 0 \\ 0 & 1 & 4 & : & 1 \\ 0 & 0 & -3 & : & 1 \end{bmatrix}, (???) \end{aligned}$$

This matrix shows that the original system has a unique solution and is reduced to

$$\begin{aligned} x_1 - x_3 &= 0 \\ x_2 + 4x_3 &= 1 \\ -3x_3 &= 1 \end{aligned}$$

The third equation gives $x_3 = -\frac{1}{3}$.

Back substituting this value in the second and first equations, we get

$$x_2 = 1 - 4x_3 = 1 + \frac{4}{3} = \frac{7}{3}, \text{ and } x_1 = x_3 = -\frac{1}{3}$$

Hence, the unique solution is $(x_1, x_2, x_3) = \left(-\frac{1}{3}, \frac{7}{3}, -\frac{1}{3}\right)$.

Example 3 Solve the following system

$$\begin{aligned} x_1 + 2x_2 - x_3 &= -2 \\ -2x_1 - 4x_2 + 2x_3 &= 4 \\ 2x_1 + 4x_2 - 2x_3 &= -4 \end{aligned}$$

Solution

The given system is

$$\begin{aligned} x_1 + 2x_2 - x_3 &= -2 \\ -2x_1 - 4x_2 + 2x_3 &= 4 \\ 2x_1 + 4x_2 - 2x_3 &= -4 \end{aligned}$$

The augmented matrix is

$$\begin{bmatrix} 1 & 2 & -1 & : & -2 \\ -2 & -4 & 2 & : & 4 \\ 2 & 4 & -2 & : & -4 \end{bmatrix}$$

Reducing it in ECHELON form, we get

$$\begin{bmatrix} 1 & 2 & -1 & : & -2 \\ -2 & -4 & 2 & : & 4 \\ 2 & 4 & -2 & : & -4 \end{bmatrix} \approx \begin{bmatrix} 1 & 2 & -1 & : & -2 \\ 0 & 0 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}, (???)$$

This matrix shows that the system has only one variable x_1 as a basic variable and the rest x_2 and x_3 as free variables. Thus, the system is consistent with infinitely many solutions.

Since x_2 and x_3 are free, they can take any value. So, let $x_2 = c$ and $x_3 = d$.

Then, the third equation implies

$$x_1 + 2c - d = -2 \Rightarrow x_1 = -2c + d - 2$$

Therefore, the required solution is

$$(x_1, x_2, x_3) = (-2c + d - 2, c, d)$$

Linear Dependence and Independence

Linearly Dependent Vectors

A set of vectors $v_1, v_2, \dots, v_n$ in C^k is said to be linearly dependent if there exists real or complex numbers $c_1, c_2, \dots, c_n$ not all zero (i.e. at least one is not zero) such that

$$c_1 v_1 + c_2 v_2 + \dots + c_n v_n = 0.$$

In other words, a set of vectors is said to be linearly dependent if one of them can be written as a linear combination of the remaining vectors.

Linearly Independent Vectors

A set of non-zero vectors $v_1, v_2, \dots, v_n$ in C^k is said to be linearly independent if for complex scalars $c_1, c_2, \dots, c_n$, the expression

$$c_1 v_1 + c_2 v_2 + \dots + c_n v_n = 0$$

is true only if $c_i = 0$ for all $i = 1, 2, \dots, n$.

Example 1 Determine whether the vectors

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 2 \\ 3 \\ 1 \\ -1 \end{bmatrix}, v_3 = \begin{bmatrix} -1 \\ 0 \\ 2 \\ 2 \end{bmatrix}, v_4 = \begin{bmatrix} 3 \\ -1 \\ 1 \\ 3 \end{bmatrix}$$

are linearly dependent or linearly independent.

Solution:

The given vectors are:

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 2 \\ 3 \\ 1 \\ -1 \end{bmatrix}, v_3 = \begin{bmatrix} -1 \\ 0 \\ 2 \\ 2 \end{bmatrix}, v_4 = \begin{bmatrix} 3 \\ -1 \\ 1 \\ 3 \end{bmatrix}$$

Let c_1, c_2, c_3, c_4 be scalars such that

$$c_1 v_1 + c_2 v_2 + c_3 v_3 + c_4 v_4 = 0$$

Or,

$$c_1 \begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 3 \\ 1 \\ -1 \end{bmatrix} + c_3 \begin{bmatrix} -1 \\ 0 \\ 2 \\ 2 \end{bmatrix} + c_4 \begin{bmatrix} 3 \\ -1 \\ 1 \\ 3 \end{bmatrix} = 0$$

Or,

$$\begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 3 & 0 & -1 \\ -1 & 1 & 2 & 1 \\ 0 & -1 & 2 & 3 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{Or, } Ax = b \quad \dots \quad (1),$$

$$\text{where, } A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 3 & 0 & -1 \\ -1 & 1 & 2 & 1 \\ 0 & -1 & 2 & 3 \end{bmatrix}, x = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} \text{ and } b = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Now the augmented matrix is

$$[A:b] = \begin{bmatrix} 1 & 2 & -1 & 3 & : & 0 \\ 2 & 3 & 0 & -1 & : & 0 \\ -1 & 1 & 2 & 1 & : & 0 \\ 0 & -1 & 2 & 3 & : & 0 \end{bmatrix}$$

$$\approx \begin{bmatrix} 1 & 2 & -1 & 3 & : & 0 \\ 0 & -1 & 2 & -7 & : & 0 \\ 0 & 3 & 1 & 4 & : & 0 \\ 0 & -1 & 2 & 3 & : & 0 \end{bmatrix}, (\text{why ?})$$

$$\approx \begin{bmatrix} 1 & 2 & -1 & 3 & : & 0 \\ 0 & -1 & 2 & -7 & : & 0 \\ 0 & 0 & 7 & -17 & : & 0 \\ 0 & 0 & 0 & 10 & : & 0 \end{bmatrix}, (\text{why ?})$$

The corresponding system of equation of this matrix is:

$$\begin{aligned} c_1 + 2c_2 - c_3 + 3c_4 &= 0 \\ -c_2 + 2c_3 - 7c_4 &= 0 \\ 7c_3 + 17c_4 &= 0 \\ 10c_4 &= 0 \end{aligned}$$

The last equation gives $c_4 = 0$ and substituting back this value, we get $c_3 = 0, c_2 = 0, c_1 = 0$. Hence, the given system is linearly independent.

Example 2 Determine whether the vector

$$v_1 = \begin{bmatrix} e^{-t} \\ 2e^{-t} \end{bmatrix}, v_2 = \begin{bmatrix} e^{-t} \\ e^{-t} \end{bmatrix}, v_3 = \begin{bmatrix} 3e^{-t} \\ 0 \end{bmatrix}$$

are linearly dependent or independent for $-\infty < t < \infty$. If they are linearly dependent find a linear relation among them.

Solution:

The given vectors are:

$$v_1 = \begin{bmatrix} e^{-t} \\ 2e^{-t} \end{bmatrix}, v_2 = \begin{bmatrix} e^{-t} \\ e^{-t} \end{bmatrix}, v_3 = \begin{bmatrix} 3e^{-t} \\ 0 \end{bmatrix}$$

Let c_1, c_2 and c_3 be scalars such that

$$\begin{aligned} c_1 v_1 + c_2 v_2 + c_3 v_3 &= 0 \\ \Rightarrow c_1 \begin{bmatrix} e^{-t} \\ 2e^{-t} \end{bmatrix} + c_2 \begin{bmatrix} e^{-t} \\ e^{-t} \end{bmatrix} + c_3 \begin{bmatrix} 3e^{-t} \\ 0 \end{bmatrix} &= 0 \\ \Rightarrow \begin{bmatrix} e^{-t} & e^{-t} & 3e^{-t} \\ 2e^{-t} & e^{-t} & 0 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \Rightarrow Ax &= b \quad \dots \quad (1) \end{aligned}$$

where,

$$A = \begin{bmatrix} e^{-t} & e^{-t} & 3e^{-t} \\ 2e^{-t} & e^{-t} & 0 \end{bmatrix}, x = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} \text{ and } b = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The augmented matrix is

$$\begin{aligned} [A:b] &= \begin{bmatrix} e^{-t} & e^{-t} & 3e^{-t} & : & 0 \\ 2e^{-t} & e^{-t} & 0 & : & 0 \end{bmatrix} \\ &\approx \begin{bmatrix} 1 & 1 & 3 & : & 0 \\ 2 & 1 & 0 & : & 0 \end{bmatrix}, (\text{why?}) \\ &\approx \begin{bmatrix} 1 & 1 & 3 & : & 0 \\ 0 & -1 & -6 & : & 0 \end{bmatrix}, (\text{why?}) \end{aligned}$$

The corresponding system of this matrix is

$$\begin{aligned} c_1 + c_2 + 3c_3 &= 0 \\ -c_2 - 6c_3 &= 0 \end{aligned}$$

This shows that the system has c_1 and c_2 as basic variables and c_3 as free variable. So, the system has infinitely many solutions.

Here c_3 is free variable and choose $c_3 = 1$.

$$\begin{aligned} \therefore -c_2 - 6c_3 &= 0 \\ \Rightarrow -c_2 - 6 &= 0 \Rightarrow c_2 = -6 \\ \text{and } c_1 + c_2 + 3c_3 &= 0 \\ \Rightarrow c_1 - 6 + 3 &= 0 \Rightarrow c_1 = 3. \end{aligned}$$

Then, the linear expression of v_1, v_2, v_3 is

$$3v_1 - 6v_2 + v_3 = 0$$

Example 3 Determine whether the vectors

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}, v_2 = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, v_4 = \begin{bmatrix} 4 \\ 3 \\ -2 \end{bmatrix}$$

are linearly dependent or linearly independent. If they are linearly dependent, find a linear relation among them.

Solution

The given vectors are

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}, v_2 = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, v_4 = \begin{bmatrix} 4 \\ 3 \\ -2 \end{bmatrix}$$

Let c_1, c_2, c_3, c_4 be scalars such that

$$\begin{aligned} c_1v_1 + c_2v_2 + c_3v_3 + c_4v_4 &= 0 \\ \Rightarrow [v_1 \ v_2 \ v_3 \ v_4] \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} &= 0 \\ \Rightarrow \begin{bmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & -1 & 3 \\ -2 & 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \end{aligned}$$

The augmented matrix is

$$\begin{bmatrix} 1 & 3 & 2 & 4 & : & 0 \\ 2 & 1 & -1 & 3 & : & 0 \\ -2 & 0 & 1 & -2 & : & 0 \end{bmatrix}$$

Reducing it into ECHELON form, we get

$$\begin{bmatrix} 1 & 3 & 2 & 4 & : & 0 \\ 2 & 1 & -1 & 3 & : & 0 \\ -2 & 0 & 1 & -2 & : & 0 \end{bmatrix}$$

$$\begin{aligned}
&\approx \begin{bmatrix} 1 & 3 & 2 & 4 & : & 0 \\ 0 & -5 & -5 & -5 & : & 0 \\ 0 & 6 & 5 & 6 & : & 0 \end{bmatrix}, (???) \\
&\approx \begin{bmatrix} 1 & 3 & 2 & 4 & : & 0 \\ 0 & 1 & 1 & 1 & : & 0 \\ 0 & 6 & 5 & 6 & : & 0 \end{bmatrix}, (???) \\
&\approx \begin{bmatrix} 1 & 0 & -1 & 1 & : & 0 \\ 0 & 1 & 1 & 1 & : & 0 \\ 0 & 0 & -1 & 0 & : & 0 \end{bmatrix}, (???) \\
&\approx \begin{bmatrix} 1 & 0 & -1 & 1 & : & 0 \\ 0 & 1 & 1 & 1 & : & 0 \\ 0 & 0 & 1 & 0 & : & 0 \end{bmatrix}, (???) \\
&\approx \begin{bmatrix} 1 & 0 & 0 & 1 & : & 0 \\ 0 & 1 & 0 & 1 & : & 0 \\ 0 & 0 & 1 & 0 & : & 0 \end{bmatrix}, (???)
\end{aligned}$$

This shows that the system has c_1, c_2 and c_3 are basic variables and c_4 is free variable. So, the system has non-trivial solutions and hence the given vectors are linearly dependent.

From the last row we see that $c_3 = 0$. And the second equation gives

$$c_2 + c_4 = 0 \Rightarrow c_2 = -c_4.$$

Also, the first row gives

$$c_1 + c_4 = 0 \Rightarrow c_1 = -c_4$$

Hence, the linear relation among v_1, v_2, v_3, v_4 is

$$-c_4 v_1 - c_4 v_2 + c_4 v_4 = 0$$

In particular, choosing $c_4 = -1$, we get

$$v_1 + v_2 - v_4 = 0.$$

Example 4 Determine whether the vectors

$$v_1 = \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}, v_2 = \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix}, v_3 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}, v_4 = \begin{bmatrix} 4 \\ 3 \\ 0 \end{bmatrix}$$

are linearly dependent or linearly independent. If they are linearly dependent, find a linear relation among them.

$$[\text{Ans: } v_1 - 5v_2 - 3v_3 + 5v_4 = 0]$$

Eigenvalues and Eigenvectors

Let A be a square matrix with real or complex entries of order n . A vector x is said to be an eigenvector of A (or a characteristics vector) if

$$Ax = \lambda x \quad \dots \quad (1)$$

for some scalar $\lambda \in \mathbb{C}$.

If $x \neq 0$, λ is uniquely determined. This λ is said to be an eigenvalue (Characteristic value) of A for the eigenvector x .

From (1), we have

$$\begin{aligned}
Ax &= \lambda x \\
\Rightarrow Ax &= \lambda Ix \\
\Rightarrow (A - \lambda I)x &= 0 \quad \dots \quad (2)
\end{aligned}$$

where I is the identity matrix of order n .

This equation (2) is now called the matrix equation, the polynomial $P_A(t) = |A - \lambda I| = \text{Det}(A - \lambda I)$ is called the characteristic polynomial of A and the equation.

$$|A - \lambda I| = 0 \quad \dots \quad (3)$$

is called characteristic equation of degree in λ . The roots of (3) are eigenvalues of the matrix A .

Note

If A is a 3×3 order matrix, equation (2) takes the form

$$\left\{ \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_{11}-\lambda & a_{12} & a_{13} \\ a_{21} & a_{22}-\lambda & a_{23} \\ a_{31} & a_{32} & a_{33}-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

And equation (3) takes the form

$$\begin{vmatrix} a_{11}-\lambda & a_{12} & a_{13} \\ a_{21} & a_{22}-\lambda & a_{23} \\ a_{31} & a_{32} & a_{33}-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (a_{11}-\lambda) \begin{vmatrix} a_{22}-\lambda & a_{23} \\ a_{32} & a_{33}-\lambda \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33}-\lambda \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22}-\lambda \\ a_{31} & a_{32} \end{vmatrix} = 0$$

Example 1 Find the eigenvalues and eigenvectors of the following matrix:

$$A = \begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix}$$

Solution:

The given matrix is

$$A = \begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix}$$

The characteristic matrix is

$$A - \lambda I = \begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3-\lambda & 2 & 4 \\ 2 & -\lambda & 2 \\ 4 & 2 & 3-\lambda \end{bmatrix}$$

The characteristic equation is

$$|A - \lambda I| = 0 \quad \dots \quad (1)$$

$$\Rightarrow \begin{vmatrix} 3-\lambda & 2 & 4 \\ 2 & -\lambda & 2 \\ 4 & 2 & 3-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (3-\lambda)(-3\lambda + \lambda^2 - 4) - 2(6 - 2\lambda - 8) + 4(4 + 4\lambda) = 0$$

$$\Rightarrow -9\lambda + 3\lambda^2 - 12 + 3\lambda^2 - \lambda^3 + 4\lambda + 4\lambda + 4 + 16 + 16\lambda = 0$$

$$\text{Or,} \quad -\lambda^3 + 6\lambda^2 + 15\lambda + 8 = 0$$

$$\text{Or,} \quad \lambda^3 - 6\lambda^2 - 15\lambda - 8 = 0$$

$$\text{Or,} \quad \lambda^3 + \lambda^2 - 7\lambda^2 - 7\lambda - 8\lambda - 8 = 0$$

$$\Rightarrow \lambda^2(\lambda + 1) - 7\lambda(\lambda + 1) - 8(\lambda + 1) = 0$$

$$\Rightarrow (\lambda + 1)(\lambda^2 - 7\lambda - 8) = 0$$

$$\Rightarrow (\lambda + 1)(\lambda^2 + \lambda - 8\lambda - 8) = 0$$

$$\Rightarrow (\lambda + 1)(\lambda + 1)(\lambda - 8) = 0$$

$$\therefore \lambda = -1, -1, 8$$

$\therefore$ the Eigenvalues of the matrix A are -1 (multiplicity 2) and 8 .

To find the eigenvectors corresponding to each Eigenvalues, we replace λ in equation (1) by the Eigenvalues.

To find the Eigenvectors Corresponding to $\lambda = -1$

Let $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ be an eigenvector(s) of A corresponding to $\lambda = -1$.

Then, the matrix equation is

$$\begin{aligned}(A - \lambda I)x &= 0 \\ \Rightarrow (A + I)x &= 0 \\ \Rightarrow \left\{ \begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} 4 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}\end{aligned}$$

The augmented matrix is $\begin{bmatrix} 4 & 2 & 4 & : & 0 \\ 2 & 1 & 2 & : & 0 \\ 4 & 2 & 4 & : & 0 \end{bmatrix} \approx \begin{bmatrix} 2 & 1 & 2 & : & 0 \\ 2 & 1 & 2 & : & 0 \\ 2 & 1 & 2 & : & 0 \end{bmatrix}$

$$\approx \begin{bmatrix} 2 & 1 & 2 & : & 0 \\ 0 & 0 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

Here, x_1 is the only basic variable and x_2 and x_3 are free variables. The system reduces to

$$2x_1 + x_2 + 2x_3 = 0$$

For particular, choose $x_2 = 2, x_3 = 1$. Then,

$$2x_1 + 2 + 2 = 0 \Rightarrow 2x_1 = -4 \Rightarrow x_1 = -2$$

Choosing $x_2 = c_1$ and $x_3 = c_2$, we have

$$\begin{aligned}2x_1 + c_1 + 2c_2 &= 0 \\ \Rightarrow 2x_1 &= -c_1 - 2c_2 \\ \Rightarrow x_1 &= -\frac{c_1}{2} - c_2\end{aligned}$$

Therefore,

$$\begin{aligned}x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} &= \begin{bmatrix} -\frac{c_1}{2} - c_2 \\ c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} -\frac{c_1}{2} - c_2 \\ c_1 + 0 \\ 0 + c_2 \end{bmatrix} = \begin{bmatrix} -\frac{c_1}{2} \\ c_1 \\ 0 \end{bmatrix} + \begin{bmatrix} -c_2 \\ 0 \\ c_2 \end{bmatrix} \\ &= c_1 \begin{bmatrix} -\frac{1}{2} \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = c_1 \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}\end{aligned}$$

Thus, the Eigenvectors corresponding to $\lambda = -1$ are

$$x^{(1)} = c_1 \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}, c_1 \neq 0, \quad x^{(2)} = c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, c_2 \neq 0$$

Eigenvector Corresponding to $\lambda = 8$

The matrix equation is

$$\begin{aligned}Ax - \lambda x &= 0 \\ \Rightarrow (A - 8I)x &= 0 \\ \Rightarrow \left\{ \begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix} - 8 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} -5 & 2 & 4 \\ 2 & -8 & 2 \\ 4 & 2 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}\end{aligned}$$

The augmented matrix is

$$\begin{aligned}
& \begin{bmatrix} -5 & 2 & 4 & : & 0 \\ 2 & -8 & 2 & : & 0 \\ 4 & 2 & -5 & : & 0 \end{bmatrix} \\
& \approx \begin{bmatrix} 2 & -8 & 2 & : & 0 \\ -5 & 2 & 4 & : & 0 \\ 4 & 2 & -5 & : & 0 \end{bmatrix} \\
& \approx \begin{bmatrix} 1 & -4 & 1 & : & 0 \\ -5 & 2 & 4 & : & 0 \\ 4 & 2 & -5 & : & 0 \end{bmatrix} \\
& \approx \begin{bmatrix} 1 & -4 & 1 & : & 0 \\ 0 & -18 & 9 & : & 0 \\ 0 & 18 & -9 & : & 0 \end{bmatrix} \\
& \approx \begin{bmatrix} 1 & -4 & 1 & : & 0 \\ 0 & -2 & 1 & : & 0 \\ 0 & 2 & -1 & : & 0 \end{bmatrix} \\
& \approx \begin{bmatrix} 1 & -4 & 1 & : & 0 \\ 0 & -2 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}
\end{aligned}$$

$\therefore$ the system has infinite solution with x_3 as free variable. Choose $x_3 = c_3$.

$\therefore$ second equation $\Rightarrow -2x_2 + c_3 = 0$

$$\Rightarrow x_2 = \frac{c_3}{2}$$

And first equation $\Rightarrow x_1 = 4x_2 - x_3 = 4 \times \frac{c_3}{2} - c_3 = c_3$

And, we have Eigenvector corresponding to $\lambda = 8$ is

$$x^{(3)} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} c_3 \\ \frac{c_3}{2} \\ c_3 \end{bmatrix} = c_3 \begin{bmatrix} 1 \\ \frac{1}{2} \\ 1 \end{bmatrix} = c_3 \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$$

Hence, the required eigenvalues and eigenvector are

$$\lambda_1 = -1, \lambda_2 = -1, \lambda_3 = 8$$

$$x^{(1)} = c_1 \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}, x^{(2)} = c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, x^{(3)} = c_3 \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$$

If we particularly choose $c_1 = c_2 = c_3 = 1$, the Eigenvectors of matrix A are

$$x^{(1)} = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}, x^{(2)} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, x^{(3)} = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}.$$

Note

Every non-zero multiple of these vectors are Eigenvectors of A .

Example 2 Find the eigenvalues and eigenvectors of the following matrix:

$$A = \begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix}$$

Solution

The given matrix is

$$A = \begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix}$$

Let λ be an Eigenvalue of A . Then the characteristic matrix is

$$(A - \lambda I) = \begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 - \lambda & i \\ -i & 1 - \lambda \end{bmatrix}.$$

The characteristic equation is

$$\begin{vmatrix} 1 - \lambda & i \\ -i & 1 - \lambda \end{vmatrix} = 0$$

$$\begin{aligned}\Rightarrow (1 - \lambda)^2 - 1 &= 0 \\ \Rightarrow \lambda^2 - 2\lambda &= 0 \\ \Rightarrow \lambda(\lambda - 2) &= 0\end{aligned}$$

Therefore, the Eigenvalues are

$$\lambda = 0, 2$$

Eigenvector corresponding to $\lambda = 0$

Let $x^{(1)} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ be the Eigenvector corresponding to $\lambda = 0$. Then the matrix equation is

$$\begin{aligned}\begin{bmatrix} 1 - 0 & i \\ -i & 1 - 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix}\end{aligned}$$

Now, the augmented matrix is

$$\begin{aligned}&\begin{bmatrix} 1 & i & : & 0 \\ -i & 1 & : & 0 \end{bmatrix} \\ &\approx \begin{bmatrix} 1 & i & : & 0 \\ 0 & 0 & : & 0 \end{bmatrix}, (???)\end{aligned}$$

This shows that x_1 is basic variable and x_2 is free variable. So, first equation gives

$$x_1 + ix_2 = 0 \Rightarrow x_1 = -ix_2.$$

Therefore, the Eigenvector corresponding to $\lambda = 0$ is

$$x^{(1)} = \begin{bmatrix} -ix_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} -i \\ 1 \end{bmatrix}$$

Eigenvector corresponding to $\lambda = 2$

Let $x^{(2)} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ be the Eigenvector corresponding to $\lambda = 2$. Then the matrix equation is

$$\begin{aligned}\begin{bmatrix} 1 - 2 & i \\ -i & 1 - 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} -1 & i \\ -i & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix}\end{aligned}$$

Now, the augmented matrix is

$$\begin{aligned}&\begin{bmatrix} -1 & i & : & 0 \\ -i & -1 & : & 0 \end{bmatrix} \\ &\approx \begin{bmatrix} 1 & -i & : & 0 \\ -i & -1 & : & 0 \end{bmatrix}, (???) \\ &\approx \begin{bmatrix} 1 & -i & : & 0 \\ 0 & 0 & : & 0 \end{bmatrix}, (???)\end{aligned}$$

This shows that x_1 is basic variable and x_2 is free variable. So, first equation gives

$$x_1 - ix_2 = 0 \Rightarrow x_1 = ix_2.$$

Therefore, the Eigenvector corresponding to $\lambda = 2$ is

$$x^{(2)} = \begin{bmatrix} ix_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} i \\ 1 \end{bmatrix}.$$

Hence, the Eigenvalues of matrix A are 0 and 2

And the Eigenvectors corresponding to them are

$$x_2 \begin{bmatrix} -i \\ 1 \end{bmatrix}, x_2 \begin{bmatrix} i \\ 1 \end{bmatrix}$$

Choosing $x_2 = 1$ particularly, the Eigenvectors are $\begin{bmatrix} -i \\ 1 \end{bmatrix}, \begin{bmatrix} i \\ 1 \end{bmatrix}$. And every non-zero scalar multiple of these vectors are also Eigenvectors of matrix A .

Example 3 Find the eigenvalues and eigenvectors of the following matrix:

$$A = \begin{bmatrix} \frac{11}{9} & -\frac{2}{9} & \frac{8}{9} \\ -\frac{2}{9} & \frac{2}{9} & \frac{10}{9} \\ \frac{8}{9} & \frac{10}{9} & \frac{5}{9} \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 11 & -2 & 8 \\ -2 & 2 & 10 \\ 8 & 10 & 5 \end{bmatrix}$$

Solution:

The given matrix is

$$A = \begin{bmatrix} \frac{11}{9} & -\frac{2}{9} & \frac{8}{9} \\ -\frac{2}{9} & \frac{2}{9} & \frac{10}{9} \\ \frac{8}{9} & \frac{10}{9} & \frac{5}{9} \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 11 & -2 & 8 \\ -2 & 2 & 10 \\ 8 & 10 & 5 \end{bmatrix}$$

The characteristic matrix is

$$\begin{aligned} A - \lambda I &= \begin{bmatrix} \frac{11}{9} & -\frac{2}{9} & \frac{8}{9} \\ -\frac{2}{9} & \frac{2}{9} & \frac{10}{9} \\ \frac{8}{9} & \frac{10}{9} & \frac{5}{9} \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \frac{11}{9} - \lambda & -\frac{2}{9} & \frac{8}{9} \\ -\frac{2}{9} & \frac{2}{9} - \lambda & \frac{10}{9} \\ \frac{8}{9} & \frac{10}{9} & \frac{5}{9} - \lambda \end{bmatrix} \end{aligned}$$

The characteristic equation is

$$\begin{aligned} |A - \lambda I| &= 0 \quad \dots \quad (1) \\ \Rightarrow \begin{vmatrix} \frac{11}{9} - \lambda & -\frac{2}{9} & \frac{8}{9} \\ -\frac{2}{9} & \frac{2}{9} - \lambda & \frac{10}{9} \\ \frac{8}{9} & \frac{10}{9} & \frac{5}{9} - \lambda \end{vmatrix} &= 0 \\ \Rightarrow \begin{vmatrix} \frac{11 - 9\lambda}{9} & -\frac{2}{9} & \frac{8}{9} \\ -\frac{2}{9} & \frac{2 - 9\lambda}{9} & \frac{10}{9} \\ \frac{8}{9} & \frac{10}{9} & \frac{5 - 9\lambda}{9} \end{vmatrix} &= 0 \\ \Rightarrow \begin{vmatrix} 11 - 9\lambda & -2 & 8 \\ -2 & 2 - 9\lambda & 10 \\ 8 & 10 & 5 - 9\lambda \end{vmatrix} &= 0 \\ \Rightarrow (11 - 9\lambda)[(2 - 9\lambda)(5 - 9\lambda) - 100] + 2[-10 + 18\lambda - 80] + 8[-20 - 16 + 72\lambda] &= 0 \\ \Rightarrow (11 - 9\lambda)[81\lambda^2 - 63\lambda - 90] + 2[18\lambda - 90] + 8[72\lambda - 36] &= 0 \\ \Rightarrow (11 - 9\lambda)[9\lambda^2 - 7\lambda - 10] + 2[2\lambda - 10] + 8[8\lambda - 4] &= 0 \\ \Rightarrow 99\lambda^2 - 77\lambda - 110 - 81\lambda^3 + 63\lambda^2 + 90 + 4\lambda - 20 + 64\lambda - 32 &= 0 \\ \Rightarrow -81\lambda^3 + 162\lambda^2 + 81\lambda - 162 &= 0 \\ \Rightarrow 81\lambda^3 - 162\lambda^2 - 81\lambda + 162 &= 0 \end{aligned}$$

$$\begin{aligned}
&\Rightarrow \lambda^3 - 2\lambda^2 - \lambda + 2 = 0 \\
&\Rightarrow \lambda^2(\lambda - 2) - (\lambda - 2) = 0 \\
&\Rightarrow (\lambda - 2)(\lambda^2 - 1) = 0 \\
&\Rightarrow (\lambda - 2)(\lambda - 1)(\lambda + 1) = 0 \\
&\therefore \lambda = 1, -1, 2
\end{aligned}$$

To find the Eigenvectors Corresponding to $\lambda = -1$

Let $x^{(1)} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ be an eigenvector(s) of A corresponding to $\lambda = -1$.

Then, the matrix equation is

$$\begin{aligned}
(A - \lambda I)x &= 0 \\
\Rightarrow (A + I)x &= 0 \\
\Rightarrow \left\{ \begin{bmatrix} \frac{11}{9} & -\frac{2}{9} & \frac{8}{9} \\ -\frac{2}{9} & \frac{2}{9} & \frac{10}{9} \\ \frac{8}{9} & \frac{10}{9} & \frac{5}{9} \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\
\Rightarrow \begin{bmatrix} \frac{20}{9} & -\frac{2}{9} & \frac{8}{9} \\ -\frac{2}{9} & \frac{11}{9} & \frac{10}{9} \\ \frac{8}{9} & \frac{10}{9} & \frac{14}{9} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}
\end{aligned}$$

The augmented matrix is

$$\begin{aligned}
&\begin{bmatrix} \frac{20}{9} & -\frac{2}{9} & \frac{8}{9} & : & 0 \\ -\frac{2}{9} & \frac{11}{9} & \frac{10}{9} & : & 0 \\ \frac{8}{9} & \frac{10}{9} & \frac{14}{9} & : & 0 \end{bmatrix} \\
&\approx \begin{bmatrix} 20 & -2 & 8 & : & 0 \\ -2 & 11 & 10 & : & 0 \\ 8 & 10 & 14 & : & 0 \end{bmatrix} \\
&\approx \begin{bmatrix} 10 & -1 & 4 & : & 0 \\ -2 & 11 & 10 & : & 0 \\ 4 & 5 & 7 & : & 0 \end{bmatrix} \\
&\approx \begin{bmatrix} -2 & 11 & 10 & : & 0 \\ 10 & -1 & 4 & : & 0 \\ 4 & 5 & 7 & : & 0 \end{bmatrix} \\
&\approx \begin{bmatrix} -2 & 11 & 10 & : & 0 \\ 0 & -56 & -56 & : & 0 \\ 0 & 27 & 27 & : & 0 \end{bmatrix}, (???) \\
&\approx \begin{bmatrix} -2 & 11 & 10 & : & 0 \\ 0 & 1 & 1 & : & 0 \\ 0 & 1 & 1 & : & 0 \end{bmatrix}, (???) \\
&\approx \begin{bmatrix} -2 & 0 & -1 & : & 0 \\ 0 & 1 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}, (???)
\end{aligned}$$

Here, x_1 and x_2 are the basic variables and x_3 is a free variable. The system reduces to

$$-2x_1 - x_3 = 0$$

$$\begin{aligned}
x_2 + x_3 &= 0 \\
x_3 &= \text{free} \\
\Rightarrow x_1 &= -\frac{1}{2}x_3 \\
x_2 &= -x_3 \\
x_3 &= \text{free}
\end{aligned}$$

Therefore, an Eigenvector corresponding to $\lambda = -1$ is

$$x^{(1)} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2}x_3 \\ -x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} -\frac{1}{2} \\ -1 \\ 1 \end{bmatrix} = c_1 \begin{bmatrix} -1 \\ -2 \\ 2 \end{bmatrix}$$

For particular, choose $c_1 = 1$. Then,

$$x^{(1)} = \begin{bmatrix} -1 \\ -2 \\ 2 \end{bmatrix}$$

To find the Eigenvectors Corresponding to $\lambda = 1$

Let $x^{(2)} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ be an eigenvector(s) of A corresponding to $\lambda = 1$.

Then, the matrix equation is

$$\begin{aligned}
(A - \lambda I)x &= 0 \\
\Rightarrow (A - I)x &= 0 \\
\Rightarrow \left\{ \begin{bmatrix} \frac{11}{9} & -\frac{2}{9} & \frac{8}{9} \\ -\frac{2}{9} & \frac{2}{9} & \frac{10}{9} \\ \frac{8}{9} & \frac{10}{9} & \frac{5}{9} \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\
\Rightarrow \begin{bmatrix} \frac{2}{9} & -\frac{2}{9} & \frac{8}{9} \\ -\frac{2}{9} & -\frac{7}{9} & \frac{10}{9} \\ \frac{8}{9} & \frac{10}{9} & -\frac{4}{9} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}
\end{aligned}$$

The augmented matrix is

$$\begin{aligned}
&\begin{bmatrix} \frac{2}{9} & -\frac{2}{9} & \frac{8}{9} & : & 0 \\ -\frac{2}{9} & -\frac{7}{9} & \frac{10}{9} & : & 0 \\ \frac{8}{9} & \frac{10}{9} & -\frac{4}{9} & : & 0 \end{bmatrix} \\
&\approx \begin{bmatrix} 2 & -2 & 8 & : & 0 \\ -2 & -7 & 10 & : & 0 \\ 8 & 10 & -4 & : & 0 \end{bmatrix} \\
&\approx \begin{bmatrix} 2 & -2 & 8 & : & 0 \\ 0 & -9 & 18 & : & 0 \\ 0 & 18 & -36 & : & 0 \end{bmatrix} \\
&\approx \begin{bmatrix} 2 & -2 & 8 & : & 0 \\ 0 & 1 & -2 & : & 0 \\ 0 & 1 & -2 & : & 0 \end{bmatrix}
\end{aligned}$$

$$\approx \begin{bmatrix} 2 & 0 & 4 & : & 0 \\ 0 & 1 & -2 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}, \quad (???)$$

$$\approx \begin{bmatrix} 1 & 0 & 2 & : & 0 \\ 0 & 1 & -2 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}, \quad (???)$$

Here, x_1 and x_2 are the basic variables and x_3 is a free variable. The system reduces to

$$x_1 + 2x_3 = 0$$

$$x_2 - 2x_3 = 0$$

$$x_3 = \text{free}$$

$$\Rightarrow x_1 = -2x_3$$

$$x_2 = 2x_3$$

$$x_3 = \text{free}$$

Therefore, an Eigenvector corresponding to $\lambda = 1$ is

$$x^{(2)} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2x_3 \\ 2x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix} = c_2 \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix}$$

For particular, choose $c_2 = 1$. Then,

$$x^{(2)} = \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix}$$

To find the Eigenvectors Corresponding to $\lambda = 2$

Let $x^{(3)} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ be an eigenvector(s) of A corresponding to $\lambda = 2$.

Then, the matrix equation is

$$(A - \lambda I)x = 0$$

$$\Rightarrow (A - 2I)x = 0$$

$$\Rightarrow \left\{ \begin{bmatrix} \frac{11}{9} & -\frac{2}{9} & \frac{8}{9} \\ -\frac{2}{9} & \frac{2}{9} & \frac{10}{9} \\ \frac{8}{9} & \frac{10}{9} & \frac{5}{9} \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -\frac{7}{9} & -\frac{2}{9} & \frac{8}{9} \\ -\frac{2}{9} & -\frac{16}{9} & \frac{10}{9} \\ \frac{8}{9} & \frac{10}{9} & -\frac{13}{9} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The augmented matrix is

$$\begin{bmatrix} -\frac{7}{9} & -\frac{2}{9} & \frac{8}{9} & : & 0 \\ -\frac{2}{9} & -\frac{16}{9} & \frac{10}{9} & : & 0 \\ \frac{8}{9} & \frac{10}{9} & -\frac{13}{9} & : & 0 \end{bmatrix}$$

$$\approx \begin{bmatrix} -7 & -2 & 8 & : & 0 \\ -2 & -16 & 10 & : & 0 \\ 8 & 10 & -13 & : & 0 \end{bmatrix}$$

$$\begin{aligned}
& \approx \begin{bmatrix} -7 & -2 & 8 & : & 0 \\ 1 & 8 & -5 & : & 0 \\ 8 & 10 & -13 & : & 0 \end{bmatrix} \\
& \approx \begin{bmatrix} 1 & 8 & -5 & : & 0 \\ -7 & -2 & 8 & : & 0 \\ 8 & 10 & -13 & : & 0 \end{bmatrix}, \quad (???) \\
& \approx \begin{bmatrix} 1 & 8 & -5 & : & 0 \\ 0 & 54 & -27 & : & 0 \\ 0 & -54 & 27 & : & 0 \end{bmatrix}, \quad (???) \\
& \approx \begin{bmatrix} 1 & 8 & -5 & : & 0 \\ 0 & 2 & -1 & : & 0 \\ 0 & -2 & 1 & : & 0 \end{bmatrix}, \quad (???) \\
& \approx \begin{bmatrix} 1 & 0 & -1 & : & 0 \\ 0 & 2 & -1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}, \quad (???)
\end{aligned}$$

Here, x_1 and x_2 are the basic variables and x_3 is a free variable. The system reduces to

$$\begin{aligned}
x_1 - x_3 &= 0 \\
2x_2 - x_3 &= 0 \\
x_3 &= \text{free} \\
\Rightarrow x_1 &= x_3 \\
x_2 &= \frac{1}{2}x_3 \\
x_3 &= \text{free}
\end{aligned}$$

Therefore, an Eigenvector corresponding to $\lambda = 1$ is

$$x^{(3)} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_3 \\ \frac{1}{2}x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 1 \\ \frac{1}{2} \\ 1 \end{bmatrix} = c_3 \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$$

For particular, choose $c_2 = 1$. Then,

$$x^{(3)} = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$$

Hence, the required eigenvalues and eigenvector are

$$\begin{aligned}
\lambda_1 &= -1, \lambda_2 = 1, \lambda_3 = 2 \\
x^{(1)} &= \begin{bmatrix} -1 \\ -2 \\ 2 \end{bmatrix}, x^{(2)} = \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix}, x^{(3)} = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}
\end{aligned}$$

Basic Theory of System of First Order Linear Differential Equations

A system of first order linear differential equations in n unknowns and n equations is

$$\left. \begin{aligned} x'_1 &= a_{11}(t)x_1 + a_{12}(t)x_2 + \dots + a_{1n}(t)x_n + f_1(t) \\ x'_2 &= a_{21}(t)x_1 + a_{22}(t)x_2 + \dots + a_{2n}(t)x_n + f_2(t) \\ &\vdots \\ x'_n &= a_{n1}(t)x_1 + a_{n2}(t)x_2 + \dots + a_{nn}(t)x_n + f_n(t) \end{aligned} \right\} \dots \quad (1)$$

This system can be written as a matrix equation as

$$\begin{bmatrix} x'_1 \\ x'_2 \\ \vdots \\ x'_n \end{bmatrix} = \begin{bmatrix} a_{11}(t) & a_{12}(t) & \cdots & a_{1n}(t) \\ a_{21}(t) & a_{22}(t) & \cdots & a_{2n}(t) \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}(t) & a_{n2}(t) & \cdots & a_{nn}(t) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} f_1(t) \\ f_2(t) \\ \vdots \\ f_n(t) \end{bmatrix}$$

$$\Rightarrow x' = A(t)x + f(t) \quad \dots \quad (2)$$

where,

$$x' = \begin{bmatrix} x'_1 \\ x'_2 \\ \vdots \\ x'_n \end{bmatrix}, A(t) = \begin{bmatrix} a_{11}(t) & a_{12}(t) & \cdots & a_{1n}(t) \\ a_{21}(t) & a_{22}(t) & \cdots & a_{2n}(t) \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}(t) & a_{n2}(t) & \cdots & a_{nn}(t) \end{bmatrix}, x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$\text{and } f(t) = \begin{bmatrix} f_1(t) \\ f_2(t) \\ \vdots \\ f_n(t) \end{bmatrix}.$$

$A(t)$ is called the coefficient matrix and $f(t)$ is called the forcing function. $A(t)$ and $f(t)$ are said to be continuous in an interval $\alpha < t < \beta$ if each of their entries is continuous in this interval.

A vector $x = d(t)$ is said to be a solution of (2), if its components satisfy the given system (1). If $f(t) = 0$ in (2), then the system is of the form

$$x' = A(t)x \quad \dots \quad (3)$$

and is called the homogeneous system.

Theorem (Statement only) (Principle of Superposition)

If $x^{(1)}$ and $x^{(2)}$ are solutions of the homogeneous system (3), then every linear combination

$$c_1 x^{(1)}(t) + c_2 x^{(2)}(t)$$

is also a solution of (3).

Note

Repeated use of the above theorem leads to the

Theorem (Statement only)

If $x^{(1)}, x^{(2)}, \dots, x^{(k)}$ are solutions of the homogeneous system (3), then every linear combination

$$c_1 x^{(1)}(t) + c_2 x^{(2)}(t) + c_3 x^{(3)}(t) + \dots + c_k x^{(k)}(t)$$

is also a solution of (3).

Wronskian

Let $x^{(1)}, x^{(2)}, \dots, x^{(n)}$ be n solutions of the homogeneous system (3).

Let $X(t) = [x^{(1)}(t) \quad x^{(2)}(t) \quad \dots \quad x^{(n)}(t)]$

$$= \begin{bmatrix} x_{11}(t) & x_{12}(t) & \cdots & x_{1n}(t) \\ x_{21}(t) & x_{22}(t) & \cdots & x_{2n}(t) \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1}(t) & x_{n2}(t) & \cdots & x_{nn}(t) \end{bmatrix}$$

Then, the determinant

$$\text{Det}(X(t)) = W[x^{(1)} \quad x^{(2)} \quad \dots \quad x^{(n)}](t)$$

is said to be the Wronskian of the solutions $x^{(1)}, x^{(2)}, \dots, x^{(n)}$. These solutions are linearly independent at some point if

$$W[x^{(1)} \quad x^{(2)} \quad \dots \quad x^{(n)}](t) \neq 0$$

Theorem (Statement Only)

If the vector functions $x^{(1)}(t)$, $x^{(2)}(t)$, ..., $x^{(n)}(t)$ are linearly independent solutions of the homogeneous system (3) for each point in the interval $\alpha < t < \beta$, then for each solution $x = \phi(t)$ of the system (3), there exist unique scalars $c_1, c_2, \dots, c_n$ such that

$$\phi(t) = c_1 x^{(1)}(t) + c_2 x^{(2)}(t) + \dots + c_n x^{(n)}(t).$$

Theorem (Statement Only)

If the vector functions $x^{(1)}(t)$, $x^{(2)}(t)$, ..., $x^{(n)}(t)$ are solutions of the homogeneous system (3) in an interval $\alpha < t < \beta$, then in this interval Wronskian $W[x^{(1)}(t) \ x^{(2)}(t) \ \dots \ x^{(n)}(t)]$ is either identically zero or never vanishes.

Note

The trace of the matrix $A(t)$ in (3) is

$\text{trace}[A(t)] = \text{sum of the diagonal entries of } A(t)$

$$= a_{11}(t) + a_{22}(t) \ \dots \ + a_{nn}(t).$$

Then the Wronskian W of $x^{(1)}(t)$, $x^{(2)}(t)$, ..., $x^{(n)}(t)$ satisfies

$$\begin{aligned} \frac{dW}{dt} &= \text{tr}(A(t))W \\ \Rightarrow \frac{dW}{W} &= \text{tr}(A(t))dt \\ \Rightarrow \ln W &= \int \text{tr}(A(t)) dt + \ln C \\ \Rightarrow \ln W - \ln C &= \int \text{tr}(A(t)) dt \\ \Rightarrow \ln \frac{W}{C} &= \int \text{tr}(A(t)) dt \\ \Rightarrow \frac{W}{C} &= e^{\int \text{tr}(A(t)) dt} \\ \Rightarrow W &= C e^{\int \text{tr}(A(t)) dt}, \end{aligned}$$

where, C is an arbitrary constant. This is known as Abel's formula.

Theorem (Statement Only)

Let $e^{(k)}$ be the standard unit vector with k^{th} component 1 and the rest zero and $x^{(1)}(t)$, $x^{(2)}(t)$, ..., $x^{(n)}(t)$ be the solutions of the homogeneous system (3) satisfying the initial conditions

$x^{(1)}(t_0) = e^{(1)}, x^{(2)}(t_0) = e^{(2)}, \dots, x^{(n)}(t_0) = e^{(n)}$ at any point t_0 in the interval $\alpha < t < \beta$, then $x^{(1)}(t)$, $x^{(2)}(t)$, ..., $x^{(n)}(t)$ form a fundamental set of solutions of the homogeneous system (3).

Example 1 Consider the linear system of homogeneous equations

$$\begin{aligned} \frac{dx_1}{dt} &= 4x_1 - 3x_2 \\ \frac{dx_2}{dt} &= 6x_1 + 5x_2 \end{aligned}$$

Then,

- Verify that $x^{(1)} = \begin{bmatrix} -e^{2t} \\ 2e^{2t} \end{bmatrix}$, $x^{(2)} = \begin{bmatrix} -e^{-t} \\ e^{-t} \end{bmatrix}$ are the solutions of the system in $-\infty < t < \infty$.
- Compute the Wronskian $W[x^{(1)} \ x^{(2)}]$ directly from definition and also check the consistency with Abel's formula

- c. Find the general solution
d. Hence solve the initial value problem

$$x' = \begin{bmatrix} -4 & -3 \\ 6 & 5 \end{bmatrix} x; \quad x(0) = \begin{bmatrix} 4 \\ -5 \end{bmatrix}.$$

Solution

The given system is

$$\begin{aligned} x_1' &= -4x_1 - 3x_2 \\ x_2' &= 6x_1 - 5x_2 \end{aligned}$$

In matrix form, this system is

$$\begin{aligned} \begin{bmatrix} x_1' \\ x_2' \end{bmatrix} &= \begin{bmatrix} -4 & -3 \\ 6 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ \Rightarrow x' &= Ax \quad \dots \quad (1) \end{aligned}$$

where, $A = \begin{bmatrix} -4 & -3 \\ 6 & -5 \end{bmatrix}$.

- a. If $x^{(1)} = \begin{bmatrix} -e^{2t} \\ 2e^{2t} \end{bmatrix}$ is a solution of (1), then

$$\begin{aligned} x &= x^{(1)} = \begin{bmatrix} -e^{2t} \\ 2e^{2t} \end{bmatrix} \\ \Rightarrow x' &= x'^{(1)} = \begin{bmatrix} -2e^{2t} \\ 4e^{2t} \end{bmatrix} \end{aligned}$$

Now,

$$Ax = \begin{bmatrix} -4 & -3 \\ 6 & 5 \end{bmatrix} \begin{bmatrix} -e^{2t} \\ 2e^{2t} \end{bmatrix} = \begin{bmatrix} 4e^{2t} - 6e^{2t} \\ -6e^{2t} + 10e^{2t} \end{bmatrix} = \begin{bmatrix} -2e^{2t} \\ 4e^{2t} \end{bmatrix} = x'$$

Hence, $x^{(1)}$ is a solution of (1).

Similarly, if $x^{(2)} = \begin{bmatrix} -e^{-t} \\ e^{-t} \end{bmatrix}$ is a solution of (1), then

$$\begin{aligned} x &= x^{(2)} = \begin{bmatrix} -e^{-t} \\ e^{-t} \end{bmatrix} \\ \Rightarrow x' &= x'^{(2)} = \begin{bmatrix} e^{-t} \\ -e^{-t} \end{bmatrix}. \end{aligned}$$

And,

$$Ax = \begin{bmatrix} -4 & -3 \\ 6 & 5 \end{bmatrix} \begin{bmatrix} -e^{-t} \\ e^{-t} \end{bmatrix} = \begin{bmatrix} 4e^{-t} - 3e^{-t} \\ -6e^{-t} + 5e^{-t} \end{bmatrix} = \begin{bmatrix} e^{-t} \\ -e^{-t} \end{bmatrix} = x'$$

Hence, $x^{(2)}$ is a solution of (1).

- b. To find Wronskian of $x^{(1)}, x^{(2)}$

$$W[x^{(1)} \quad x^{(2)}] = \begin{vmatrix} -e^{2t} & -e^{-t} \\ 2e^{2t} & e^{-t} \end{vmatrix} = e^t$$

Next, $A = \begin{bmatrix} -4 & -3 \\ 6 & 5 \end{bmatrix}$.

$$\therefore \text{tr}(A) = -4 + 5 = 1.$$

If t_0 is any given real number, then, by Abel's formula

$$W(t) = W(t_0)e^{\int_{t_0}^t \text{tr}(A)dt} = e^{t_0} \times e^{\int_{t_0}^t dt} = e^{t_0} \times e^{t-t_0} = e^t$$

- c. Since

$W(t) = e^t \neq 0 \forall t \in (-\infty, \infty)$, $\{x^{(1)}, x^{(2)}\}$ is a fundamental set of solutions of (1). So the general solution is

$$x = c_1 x^{(1)} + c_2 x^{(2)}$$

$$\begin{aligned}
&= c_1 \begin{bmatrix} -e^{2t} \\ 2e^{2t} \end{bmatrix} + c_2 \begin{bmatrix} -e^{-t} \\ e^{-t} \end{bmatrix} \\
&= \begin{bmatrix} -e^{2t} & -e^{-t} \\ 2e^{2t} & e^{-t} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \\
\therefore x &= \begin{bmatrix} -e^{2t} & -e^{-t} \\ 2e^{2t} & e^{-t} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}.
\end{aligned}$$

d. The given initial condition is $x(0) = \begin{bmatrix} 4 \\ -5 \end{bmatrix}$.

Using it in the general solution we get

$$\begin{aligned}
\begin{bmatrix} 4 \\ -5 \end{bmatrix} &= \begin{bmatrix} -e^{2t} & -e^{-t} \\ 2e^{2t} & e^{-t} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \\
\Rightarrow \begin{bmatrix} 4 \\ -5 \end{bmatrix} &= \begin{bmatrix} -1 & -1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}, \text{ using } t = 0
\end{aligned}$$

The augmented matrix is

$$\begin{aligned}
&\begin{bmatrix} -1 & -1 & : & 4 \\ 2 & 1 & : & -5 \end{bmatrix} \\
&\approx \begin{bmatrix} 1 & 1 & : & -4 \\ 2 & 1 & : & -5 \end{bmatrix} \\
&\approx \begin{bmatrix} 1 & 1 & : & -4 \\ 0 & -1 & : & 3 \end{bmatrix} \\
&\approx \begin{bmatrix} 1 & 1 & : & -4 \\ 0 & 1 & : & -3 \end{bmatrix} \\
&\approx \begin{bmatrix} 1 & 0 & : & -1 \\ 0 & 1 & : & -3 \end{bmatrix}
\end{aligned}$$

Therefore, $c_1 = -1, c_2 = -3$.

Therefore, the particular solution is

$$\begin{aligned}
x &= -1 \cdot \begin{bmatrix} -e^{2t} \\ 2e^{2t} \end{bmatrix} - 3 \cdot \begin{bmatrix} -e^{-t} \\ e^{-t} \end{bmatrix} \\
x &= \begin{bmatrix} e^{2t} \\ -2e^{2t} \end{bmatrix} + \begin{bmatrix} 3e^{-t} \\ -3e^{-t} \end{bmatrix} \\
x &= \begin{bmatrix} e^{2t} + 3e^{-t} \\ -2e^{2t} - 3e^{-t} \end{bmatrix}
\end{aligned}$$

Exercise Consider $A = \begin{bmatrix} 2 & 4 \\ 4 & 2 \end{bmatrix}$, $x^{(1)} = \begin{bmatrix} e^{6t} \\ e^{6t} \end{bmatrix}$, $x^{(2)} = \begin{bmatrix} e^{-2t} \\ -e^{-2t} \end{bmatrix}$ and $k = \begin{bmatrix} -3 \\ 9 \end{bmatrix}$.

i) Prove that $\{x^{(1)}, x^{(2)}\}$ is a fundamental set of solutions of the homogeneous system

$$x' = Ax \quad \dots \quad (1)$$

ii) Solve the initial value problem

$$x' = Ax; x(0) = k$$

iii) from (ii), obtain the generalized formula for the solution (1) for an arbitrary initial value k .